Practice Set Module -2

<u>Probability Distribution.</u>

1) For the following probability distribution find $E(X), E(X2\ ), E(X-\mu)^2$

X	8	12	16	20	24
P(X=x)	1/8	1/6	3/8	1/4	1/12

2) A Continuous random variable X has the p.d.f. $f(x) = kx^2e^{-x}, x \geq 0$. Find k, mean and variance.
3) The Probability that a bomb dropped from a plane will strike the target is 1/5. If six such bombs are dropped, find the probability that (i) exactly two bombs hit the target, (ii)at least two will hit the target.
4) Take 100 sets of 10 tosses of an unbiased coin. In how many cases do you expect to get (a) 7 heads and 3 tails, (b) 7 heads at least?
5) Fit a binomial Distribution to the following data.

X	0	1	2	3	4	5	6
P(X=x)	5	18	28	12	7	6	4

6) Which Probability distribution is appropriate to describe the situation where 100 misprints are randomly distributed over 100 pages of a book. For this distribution find the probability that a page selected at random will contain atleast 3 misprints?
7) Fit a Poisson distribution to the following data.

X	0	1	2	3	4	5
P(X=x)	142	156	69	27	5	1

8) Of a large group of mean 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming a normal distribution, find the mean and standard deviation of the distribution.
9) The life of Army shoes is normally distributed with mean 8 months and SD 2 months. If 5000 pairs are issued, how many pairs would be expected to need replacement after 12 months?
10) The daily sales of a certain item are normally distributed with mean Rs8000 and SD Rs12000.
    1] What is the probability that in a certain day the sales will be less than Rs8210?
    2] What % of the days will the sales be between Rs8100 & Rs8210?